

Crude to Profit engineering manual

Estimate annual product sales, crude purchases, gross margin and margin after configured operating costs for the selected feed, technologies and prices.

Run the Python model

Open `crude_to_profit_engine.py` in Python IDLE. Select Run, then Run Module (F5). Python 3.10 or newer is required; no extra packages or other files are needed.

The built-in example writes a results JSON file beside the Python file. Edit SETTINGS near the top and run again to change assumptions. To use your own measurements, run the file with an input JSON filename. Inputs use the fields and units below.

The input envelope contains `schemaVersion`, `model`, `source`, `timezone`, `config` and `measurements`. Each measurement contains `timestamp`, `values` and optional `quality`. Use ISO timestamps with an offset. Bad, stale and missing signals remain unavailable. The AVEVA mapping is provisional; a live connection has not been verified.

Inputs and units

Model inputs · 5 rows		
INPUT	UNIT	USE
Crude feed rates	m³/h	Blend the five workbook crudes.
Residue technology	None / LC Finer / Delayed Coker	Treat vacuum residue or retain it unpriced.
Gas-oil technology	None / Hydrocracker / FCC	Treat straight-run and eligible converted gas oils once.
Price case	Low / High / Live market	C\$/m³ feed and product prices; dated sources appear in Price snapshot.
Yields, densities, cut points and coker carbon residue	Stored assumptions	Convert feeds into product volumes; assumptions are not live plant measurements.

Calculations

PRIMARY SPLIT

Cut flow = crude flow × volume yield / 100

Calculate blend shares, barrel flow and sulfur mass. VGO is the sum of its light, medium and heavy cuts.

RESIDUE CONVERSION

Product mass = residue mass × weight yield / 100; volume = mass / density

LC Finer uses the workbook split. Delayed coking uses feed carbon residue (CCR).

DELAYED COKER

Gas = $7.80 + 0.144\text{CCR}$; naphtha = $11.29 + 0.343\text{CCR}$; coke = 1.60CCR

wt%. Gas oil is the remainder to 100%; it remains one pool rather than invented diesel/VGO subcuts.

GAS-OIL CONVERSION

Pool = straight-run VGO + converted gas oil + heavy light-stream tails

Hydrocracker uses the workbook volume yields. FCC uses the Grace 75 wt% conversion case; vacuum residue does not feed FCC directly.

ECONOMICS

Hourly margin = $\Sigma(\text{product volume} \times \text{price}) - \Sigma(\text{crude volume} \times \text{price})$

Annual gross margin = hourly margin × 24 × 330 / 1,000,000 in million C\$/year. Gross margin is retained separately.

OPERATING COSTS

Annual OPEX = annual sales revenue × OPEX percentage / 100

Editable default: 8% for each price case. A single allowance covers operating costs including utilities, chemicals, maintenance and overhead. Earlier saved itemized costs remain active until replaced with the percentage.

MARGIN AFTER CONFIGURED OPEX

Annual margin after OPEX = annual sales – annual crude purchases – annual configured OPEX

Operating margin before depreciation, financing, tax and capital expenditure. The percentage replaces itemized OPEX rather than adding to it. Crude purchases remain separate.

Useful trends

- Reconcile sales minus crude purchases minus configured OPEX for each price case.
- Compare Low, High and Live market annual estimates for the same feed and routing.
- Compare product volumes and sales contribution across technology choices.

- Review price observation dates, cached sources and pricing proxies before comparing scenarios. The page is a scenario calculator, not a historical revenue forecast.
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Assumptions and limits

- Four technologies form two independent selections, giving nine routing combinations including bypass. Feed is counted once and no inter-unit recycle is modelled.
- Grace FCC yields are 2.2% dry gas, 13.3% LPG, 51.9% gasoline, 16.7% LCO, 8.6% bottoms and 7.1% coke. The reported 0.2% gap is unallocated, not saleable product.
- Client product headings are retained: FCC gasoline uses Naphtha, LCO uses Diesel, and bottoms uses UCO pricing proxies. These are not certified finished products.
- Coke, dry gas and held residue have no sales credit. The total operating allowance is deducted once after gross margin. Capital, depreciation, financing and taxes are excluded. Fixed yields and densities do not replace a refinery simulation.
- Low and High pair their respective crude and product prices; they are not guaranteed worst/best margins. Live uses the latest published observations and derived prices, not executable quotes.
- The data were obtained from open resources. Client workbook assumptions and separately sourced yield cases are identified above.

Measured values come from instruments or samples. Calculated values use those inputs and the configured assumptions. Design values are references. Simulated values demonstrate the model and are not plant measurements. Missing results remain unavailable.

Sources

- Suncor 2025: R&M operating, selling and general expenses CAD 2,439 million / operating revenue CAD 30,671 million = 7.95%, rounded to an 8% planning allowance. Includes marketing/general costs; not a technology-specific utility estimate.
- Client Crude to Profit Rev 1 workbook: feed, product headings, fixed prices and annualization
- Grace: Strategies for Maximizing FCC Light Cycle Oil, Table 1, printed p.48
- Price snapshot: observation dates and individual pricing sources